Dividing by Two-Digit Numbers

Find easy parts of the divisor. Take them out. Add.

Easy parts of 24

$24 \times 10 = 240$

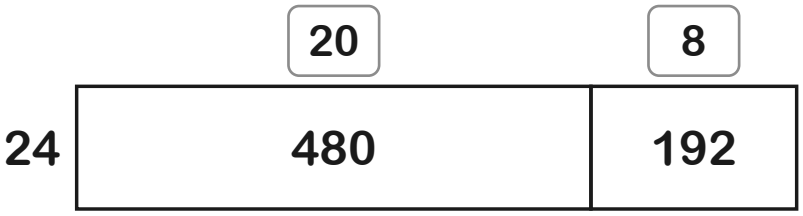
$24 \times 5 = 120$

$24 \times 20 = 480$

$24 \times 2 = 48$

Times 10 is easy.
Double it for 20.
Half of 10 is 5.

672 ÷ 24



$480 \div 24 = 20$

$192 \div 24 = 8$

$20 + 8 = 28$

Take out 20 groups of 24 (480). 192 is left. 192 is 8 groups of 24.

Not even? Write what is left

$500 \div 24 = 20 \text{ R } 20$

20 groups of 24 is 480. 20 are left, and 20 is less than 24.

Your turn!

$384 \div 16 = \boxed{24}$

For teachers

Goal: divide by two-digit numbers with the area model and partial quotients (5.NBT.6), like packet 59 with a bigger side. Say "What easy part of 24 fits? 10 groups? 20 groups?" Any easy parts work; fewer parts is faster.

Dividing by Two-Digit Numbers: Practice

Easy parts first: 10 groups, 20 groups, 5 groups.

Part 1 Easy parts Times 10, times 20, times 5.

1) $32 \times 10 = 320$	2) $17 \times 20 = 340$	3) $47 \times 5 = 235$
4) $23 \times 30 = 690$	5) $41 \times 10 = 410$	6) $28 \times 5 = 140$
7) $19 \times 20 = 380$	8) $36 \times 10 = 360$	9) $14 \times 30 = 420$
10) $52 \times 5 = 260$	11) $27 \times 20 = 540$	12) $38 \times 10 = 380$

Dividing by Two-Digit Numbers: Practice (continued)

Part 2 The area model The parts are inside. Find each side on top.

13) $714 \div 21 = 34$

	30	4
21	630	84

30 + 4 = 34

14) $832 \div 32 = 26$

	20	6
32	640	192

20 + 6 = 26

15) $705 \div 15 = 47$

	40	7
15	600	105

40 + 7 = 47

Dividing by Two-Digit Numbers: Practice (continued)

Part 3 Pick your own parts Write the parts inside, then each side on top.

16) $910 \div 26 = 35$

	30	5
26	780	130

30 + 5 = 35

17) $903 \div 43 = 21$

	20	1
43	860	43

20 + 1 = 21

18) $972 \div 18 = 54$

	50	4
18	900	72

50 + 4 = 54

Dividing by Two-Digit Numbers: Practice (continued)

Part 4 Divide each part Then add.

19) $925 = 750 + 175$

$750 \div 25 = 30$

$175 \div 25 = 7$

$30 + 7 = 37$

20) $744 = 620 + 124$

$620 \div 31 = 20$

$124 \div 31 = 4$

$20 + 4 = 24$

21) $928 = 800 + 128$

$800 \div 16 = 50$

$128 \div 16 = 8$

$50 + 8 = 58$

22) $966 = 840 + 126$

$840 \div 42 = 20$

$126 \div 42 = 3$

$20 + 3 = 23$

23) $988 = 910 + 78$

$910 \div 13 = 70$

$78 \div 13 = 6$

$70 + 6 = 76$

24) $986 = 680 + 306$

$680 \div 34 = 20$

$306 \div 34 = 9$

$20 + 9 = 29$

Part 5 Divide No help this time.

25) $672 \div 21 = 32$

26) $644 \div 14 = 46$

27) $594 \div 33 = 18$

28) $624 \div 26 = 24$

29) $663 \div 17 = 39$

30) $720 \div 45 = 16$

Dividing by Two-Digit Numbers: Practice (continued)

Part 5 Divide (continued)

31) $876 \div 12 = 73$

32) $798 \div 38 = 21$

33) $621 \div 23 = 27$

34) $714 \div 51 = 14$

35) $836 \div 19 = 44$

36) $792 \div 36 = 22$

37) $868 \div 28 = 31$

38) $870 \div 15 = 58$

39) $836 \div 44 = 19$

Part 6 Bigger numbers No help this time.

40) $5{,}544 \div 24 = 231$

41) $5{,}356 \div 13 = 412$

Dividing by Two-Digit Numbers: Practice (continued)

Part 6 Bigger numbers (continued)

42) $4{,}992 \div 32 = 156$

43) $7{,}245 \div 21 = 345$

44) $6{,}016 \div 47 = 128$

45) $6{,}000 \div 16 = 375$

46) $7{,}245 \div 35 = 207$

47) $4{,}752 \div 18 = 264$

48) $5{,}211 \div 27 = 193$

49) $7{,}436 \div 52 = 143$

Part 7 What is left? No help this time.

50) $500 \div 23 = 21$ R 17

51) $743 \div 32 = 23$ R 7

52) $618 \div 25 = 24$ R 18

53) $1{,}000 \div 37 = 27$ R 1

Dividing by Two-Digit Numbers: Practice (continued)

Part 7 What is left? (continued)

54) $859 \div 41 = 20$ R 39

55) $432 \div 19 = 22$ R 14

56) $2{,}015 \div 46 = 43$ R 37

57) $967 \div 28 = 34$ R 15

Part 8 Divide

58) $836 \div 38 = 22$

59) $299 \div 13 = 23$

60) $2{,}880 \div 45 = 64$

61) $1{,}664 \div 26 = 64$

62) $2{,}646 \div 49 = 54$

63) $1{,}518 \div 33 = 46$

Dividing by Two-Digit Numbers: Practice (continued)

Part 8 Divide (continued)

64) $988 \div 26 = 38$

65) $1{,}176 \div 21 = 56$

66) $798 \div 19 = 42$

67) $1{,}984 \div 31 = 64$

68) $323 \div 17 = 19$

69) $1{,}023 \div 33 = 31$

70) $625 \div 25 = 25$

71) $1{,}128 \div 24 = 47$

72) $384 \div 24 = 16$

73) $1{,}120 \div 40 = 28$

74) $950 \div 25 = 38$

75) $1{,}056 \div 44 = 24$

Dividing by Two-Digit Numbers: Practice (continued)

Part 8 Divide (continued)

76) $832 \div 32 = 26$	77) $1{,}833 \div 47 = 39$	78) $925 \div 25 = 37$
79) $364 \div 13 = 28$	80) $486 \div 18 = 27$	81) $950 \div 19 = 50$
82) $1{,}178 \div 31 = 38$	83) $504 \div 18 = 28$	84) $1{,}680 \div 48 = 35$
85) $828 \div 18 = 46$	86) $2{,}976 \div 48 = 62$	87) $1{,}292 \div 19 = 68$

Dividing by Two-Digit Numbers: Practice (continued)

Part 8 Divide (continued)

88) $2{,}842 \div 49 = 58$

89) $1{,}062 \div 18 = 59$

90) $546 \div 14 = 39$

91) $1{,}173 \div 23 = 51$

92) $391 \div 17 = 23$

93) $1{,}340 \div 20 = 67$

Part 9 Remember? Problems from earlier packets.

94) $324 \times 27 = 8{,}748$

95) $5.6 + 3.85 = 9.45$

96) $7 - 2.36 = 4.64$

97) $1.807 > 1.78$

98) $0.006 = 6$ thousandths

99) $4.5 \times 100 = 450$

Dividing by Two-Digit Numbers: Practice (continued)

Part 10 Word problems How many groups? How many in each?

100) A school has 648 kids in 24 equal classes. How many kids are in each class?

Number sentence **648 ÷ 24 = 27**

Answer **27** kids

Draw or show your work

101) A farmer packs 936 eggs in cartons of 12. How many cartons does he fill?

Number sentence **936 ÷ 12 = 78**

Answer **78** cartons

Draw or show your work

102) A bus trip is 1,350 miles. The bus drives 45 miles each hour. How many hours is the trip?

Number sentence **1,350 ÷ 45 = 30**

Answer **30** hours

Draw or show your work

103) A teacher has 500 stickers to share among 32 kids. How many does each kid get?

Number sentence **500 ÷ 32 = 15 R 20**

Answer **15** stickers

Draw or show your work

104) Challenge: 400 kids go on a trip. A bus holds 36 kids. How many buses do they need?

Number sentence **400 ÷ 36 = 11 R 4, so one more bus**

Answer **12** buses

Draw or show your work

Check-In: Dividing by Two-Digit Numbers

Show what you know. No help this time.

Mastery: 8 of 10 (80%)

Part 1 Divide each part

1) $792 = 660 + 132$

$660 \div 22 = 30$

$132 \div 22 = 6$

$30 + 6 = 36$

2) $731 = 680 + 51$

$680 \div 17 = 40$

$51 \div 17 = 3$

$40 + 3 = 43$

Part 2 Divide

3) $782 \div 23 = 34$

4) $630 \div 42 = 15$

5) $742 \div 14 = 53$

6) $840 \div 35 = 24$

7) $837 \div 31 = 27$

8) $702 \div 27 = 26$

Part 3 What is left?

9) $915 \div 42 = 21$ R 33

10) $621 \div 17 = 36$ R 9